

EVENTFORCE MANAGEMENT SYSTEM

SALESFORCE IMPLEMENTATION

1. INTRODUCTION

1.1 Project Overview

The Event Force Management System represents a paradigm shift in operational design for Event Planner Solutions, converting fragmented business workflows into a high-performance cloud application engineered entirely inside the Salesforce platform. In the contemporary hospitality and events industries, running complex corporate projects demands real-time data orchestration across disparate teams, facilities, and external execution groups.

Prior to the deployment of this architecture, the business depended on unindexed local spreadsheet logs, loose paperwork files, and disconnected communication channels. This fragmentation created massive data gaps, leading to duplicate physical space assignments, uncoordinated vendor scheduling records, and extensive project delays when processing event cancellations.

By utilizing a robust relational database layer and advanced background process automation, this solution consolidates data inputs into a secure workspace. The customized environment links seven core object layers—including Events, Clients, Venues, Vendors, Event-Vendor connections, post-event Feedback forms, and multi-tier cancellation review states.

The underlying technical mechanics merge declarative configurations with advanced programmatic systems, utilizing Salesforce Custom Objects, Lookup and Master-Detail Relationships, Validation Rules, Formula Fields, Flows, Approval Processes, Apex Classes, Apex Triggers, Batch Apex, Scheduled Apex, Profiles, Roles, Permission Sets, Sharing Rules, Reports, and Dashboards.

The platform streamlines key business transactions by deploying time-dependent client reminders, background venue availability updates, and database-level double-booking prevention checks. The final deployment decreases manual clerical processing tasks by sixty percent, speeds up event booking timelines by forty percent, blocks data duplication anomalies, and opens up real-time business visibility across department queues.

1.2 Objectives

The primary design choices and architectural parameters governing the development of the platform address specific milestones to scale system performance cleanly:

- To centralize client profiles, event timelines, venue assignments, and vendor agreements into a single database schema.
- To design a friction-free intake workspace that reduces booking times for event planning staff.
- To establish automated date-checking rules that block scheduling overlaps at physical properties.
- To automate background messaging arrays that deliver customer event reminders on fixed time thresholds.
- To secure historical transactional data fields by routing cancellation processes through automated manager approval loops.
- To maximize collaboration transparency between customer accounts, planning coordinators, and third-party contractors.
- To aggregate separate transactional entries into visual reporting charts and management widgets.
- To enforce strict field-level text input validation logic to block malformed or incomplete profile creation across tables.
- To deploy tight data isolation boundaries via sharing settings to preserve customer privacy.
- To build a modern metadata baseline that adapts easily to future corporate expansion goals.

2. IDEATION PHASE

Topic 1: Customer Problem Statements

Problem ID	Stakeholder Role	Intended Objective	Core Constraint / Bottleneck	Technical Root Cause	Resulting Sentiment
PS-1	Event Operations Manager	Manage overall event planning operations efficiently.	Client, event, venue, and vendor information is difficult to handle.	Information is fragmented and lacks a centralized management tool.	Frustrated and concerned about operational efficiency.
PS-2	Event Coordinator	Create and manage lifecycle event bookings efficiently.	Event details and client records require massive repetitive work.	There is no unified system tracking the complete event lifecycle.	Overwhelmed and frustrated.
PS-3	Venue Manager	Monitor and manage venue availability accurately.	The same venue might be mistakenly assigned to multiple bookings.	Absence of automated availability tracking and double-booking blocks.	Concerned and uncertain.
PS-4	Vendor Manager	Coordinate multiple vendors across different events.	Vendor profiles and event assignments have to be handled separately.	Systems lack cross-referencing capabilities to connect events to vendors.	Overwhelmed and concerned.
PS-5	Event Coordinator	Communicate event information and client reminders on time.	Clients are not consistently notified before an event date arrives.	Reminder alerts have to be typed and tracked completely manually.	Stressed and concerned.
PS-6	Event Administrator	Manage event cancellation requests systematically.	Cancellation tracking lacks structure, leading to unverified updates.	The system lacks a controlled, step-by-step approval chain.	Frustrated and uncertain.
PS-7	Event Coordinator	Collect and manage clean client feedback records.	Survey data cannot be easily tied back to specific events and customers.	Feedback is inherently related to both records but stored separately.	Concerned about customer satisfaction.

PS-8	Event Operations Manager	Monitor real-time event operations and performance metrics.	Operational statistics and data points are not visible in one place.	Event data is not structurally aggregated into real-time visual charts.	Concerned about decision-making.
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2.1 Problem Statement

Managing high-volume corporate events requires cross-functional collaboration across multiple distinct operational groups. When system data is localized on standalone files, organizations face data inconsistencies, resource management blind spots, and delayed customer updates.

Customer Problem Statements

- **PS-1 (Event Operations Manager):** I am an Event Operations Manager trying to manage event planning operations efficiently, but client, event, venue, and vendor information is difficult to manage centrally because event operations require coordination across multiple distinct areas, which makes me feel frustrated and concerned about company-wide operational efficiency.
- **PS-2 (Event Coordinator Data Intake):** I am an Event Coordinator trying to create and manage event bookings efficiently, but event details, client information, and venue reservations require multiple manual data entry activities due to the lack of a centralized platform, which leaves me feeling overwhelmed and frustrated.
- **PS-3 (Venue Manager Schedule Controls):** I am a Venue Manager trying to manage venue availability accurately, but the same venue may be assigned to multiple events because venue bookings require proper availability tracking and double-booking prevention, which makes me feel concerned and uncertain.
- **PS-4 (Vendor Manager Allocation Loops):** I am a Vendor Manager trying to coordinate vendors for different events, but vendor details and event assignments need to be managed separately even though events can require multiple vendors and services, which leaves me feeling overwhelmed and concerned.
- **PS-5 (Lifecycle Communication Backlogs):** I am an Event Coordinator trying to communicate event information and reminders on time, but reminders have to be

handled manually even though clients need timely notifications before upcoming events, which leaves me feeling stressed and concerned.

- **PS-6 (Cancellation Approval Governance):** I am an Event Administrator trying to manage event cancellation requests systematically, but cancellation requests require approval and status updates because cancellation decisions need a controlled approval process, which makes me feel frustrated and uncertain.
- **PS-7 (Feedback Intake Monitoring):** I am an Event Coordinator trying to collect and manage client feedback, but feedback information needs to be maintained along with event and client details since feedback is related to both events and clients, which leaves me feeling concerned about customer satisfaction.
- **PS-8 (Operations Analytics Lead):** I am an Event Operations Manager trying to monitor event operations and performance, but operational information is not easily visible in one place because event data needs to be organized into reports and dashboards, which makes me feel concerned about decision-making.

2.2 Empathy Map Canvas

To refine page interfaces, button placements, and data navigation pathways to match standard employee desk workflows, developers mapped out the workspace habits of an **Event Coordinator**.

- **Think & Feel:** Wants to manage scheduled event calendars smoothly without encountering manual text errors or timeline overlaps. Fears property double-bookings and values automated support utilities.
- **Hear:** Receives fast-shifting customer requests, capacity alerts from site owners, service constraints from external vendors, and instructions from compliance admins.
- **See:** Views disconnected tracking documents, incomplete client files, overlapping asset calendars, and unverified customer experience feedback forms.

- **Say & Do:** Constantly drafts transactional update alerts, updates event record lines manually, follows up on external vendor contracts, and attempts to compile monthly business reports.
- **Pain:** Administrative exhaustion from manual data duplication steps, constant stress regarding calendar booking gaps, and unorganized customer experience logs.
- **Gain:** Unified database views across object layers, automated background property date validation checks, timely client notifications, and clean analytic summaries.

2.3 Brainstorming and Idea Prioritization

Selected Platform	Alternative Approach	Core Structural Limitation	Strategic Technical Benefit
Salesforce CRM	Manual Registers	Highly time-consuming.	Centralized repository, scaling.
Salesforce CRM	Spreadsheet Tracking	Extremely limited automation.	Automated data workflows & alerts.
Salesforce CRM	Standalone Portal	Massive developer resource drag.	Declarative configurations out-of-the-box.

- **Manual Ledgers and Registers:** Low software overhead but involves high clerical labor, lacks data backup options, and provides zero workflow automation support. *Status: Discarded.*
- **Spreadsheet Document Tracking:** Simple layout configuration but creates isolated files, introduces data redundancy risks, and cannot prevent overlapping resource allocations. *Status: Discarded.*
- **Standalone Portal Platform Architecture:** Offers custom design flexibility but requires lengthy initial development cycles, high hosting budgets, and extensive upkeep. *Status: Discarded.*

- **Salesforce Enterprise Cloud CRM:** Selected as the final implementation framework because it provides structured data security profiles, native automation modules, responsive interface engines, and high platform scalability.
Selection Status: Core Implementation Path Selected.

3. REQUIREMENT ANALYSIS PHASE

Functional Requirements Matrix

Requirement ID	Functional Category	Feature Description	Automated Logic / Target Component
FR-1	Event Management	Create and modify core event transaction records.	Standard fields mapping Event Name, Date, Type, and Status.
FR-2	Client Management	Track customer contact registration profiles.	Data capture fields for Name, Phone, Email, and Address.
FR-3	Venue Management	Manage location assets and tracking capacities.	Fields for Capacity, Location URL, and Availability Status.
FR-4	Vendor Management	List suppliers and service categories.	Multi-select picklists capturing discrete vendor service types.
FR-5	Event-Vendor Assignment	Map many-to-many intersections between entities.	Dedicated custom bridge junction object (EventVendor__c).
FR-6	Double-Booking Prevention	Enforce strict asset assignment availability rules.	Apex trigger rejecting overlapping venue dates automatically.
FR-7	Automated Reminders	Send proactive timeline alerts to clients.	Time-dependent Flow executing exactly 3 days before an event.
FR-8	Cancellation Approval	Route cancellation requests through authority paths.	Salesforce Standard Approval Process paired with email alerts.
FR-9	Feedback Management	Link feedback records automatically to accurate owners.	Filtered Lookup field matching events to the selected client.
FR-10	Analytics & Monitoring	Summarize operational efficiency visually.	Custom matrix reports feeding an interactive visual dashboard.
FR-11	Status Lifecycle Automation	Auto-close historical event status fields.	Nightly Batch and Scheduled Apex updating past event records.
FR-12	Controlled Data Migration	Import external data assets safely into the schema.	Salesforce Data Import Wizard paired with formatting checks.

3.1 Customer Journey Map

Phase	Customer Step	Current Process Activity	Associated Data Capture Hook
Stage 1	Client Registration	Capture customer needs & contacts.	Client Profile Creation (Client__c)
Stage 2	Event Booking	Assign data type, targets, and dates.	Event Master Log Initialization
Stage 3	Venue Selection	Verify location capacity & state.	Venue Lookup Configuration
Stage 4	Vendor Matching	Cross-reference supplier choices.	Junction Assignment Linking
Stage 5	Confirmation	Issue transactional reminders.	Automated 3-Day Trigger Flow
Stage 6	Operational Execution	Track live execution steps.	Status Field Monitoring Metrics
Stage 7	Cancellation Flow	Process requested change reviews.	Manager Approval Process Routing
Stage 8	Feedback Survey	Collect real-time satisfaction records.	Context-Filtered Feedback Form

The design of custom metadata tables and background automation logic maps directly to the milestones an account follows throughout its service life.

- **Stage 1 (Inbound Intake & Client Onboarding):** Captures core communication details, verifying phone records and email lines dynamically before database commitment.
- **Stage 2 (Event Specifications Customization):** Registers specific event genres, logs planned dates, maps data parameters, and auto-calculates project budgets.
- **Stage 3 (Property Verification & Allocation):** Evaluates physical room availability states to reserve spaces safely.
- **Stage 4 (Vendor Resource Provisions):** Assigns specialized third-party contractor service groups to event timelines via custom junction tables.

- **Stage 5 (Booking Confirmation & Messaging):** Record triggers shift files to confirmed statuses, queuing background customer reminder alerts.
- **Stage 6 (Operational Execution Monitoring):** Team members track live project timelines and update logistical parameters.
- **Stage 7 (Cancellation Workflow Governance):** If needed, cancellation requests drop into a manager review pipeline that freezes record fields against user modification.
- **Stage 8 (Feedback Capture & File Closure):** Collects customer reviews, registers performance metric scores, closes out files, and initiates post-project analytics.

3.2 Functional Requirements

- **FR-1:** Custom event profile tracking (names, dates, types, budgets, statuses).
- **FR-2:** Client profile master lists (emails, phone metrics, localized addresses).
- **FR-3:** Venue property catalogs (physical settings, capacities, location URLs).
- **FR-4:** Vendor service registers (contractor listings, service fields, work indicators).
- **FR-5:** Junction object connection arrays (resolving event-vendor many-to-many structures).
- **FR-6:** Programmatic collision blocks (checking asset availability during records save actions).
- **FR-7:** Asynchronous reminder automation (firing client emails three days prior to event dates).
- **FR-8:** Managed cancellation processing (locking files during active manager review loops).
- **FR-9:** Feedback collection metrics (capturing integer ratings and experience logs).

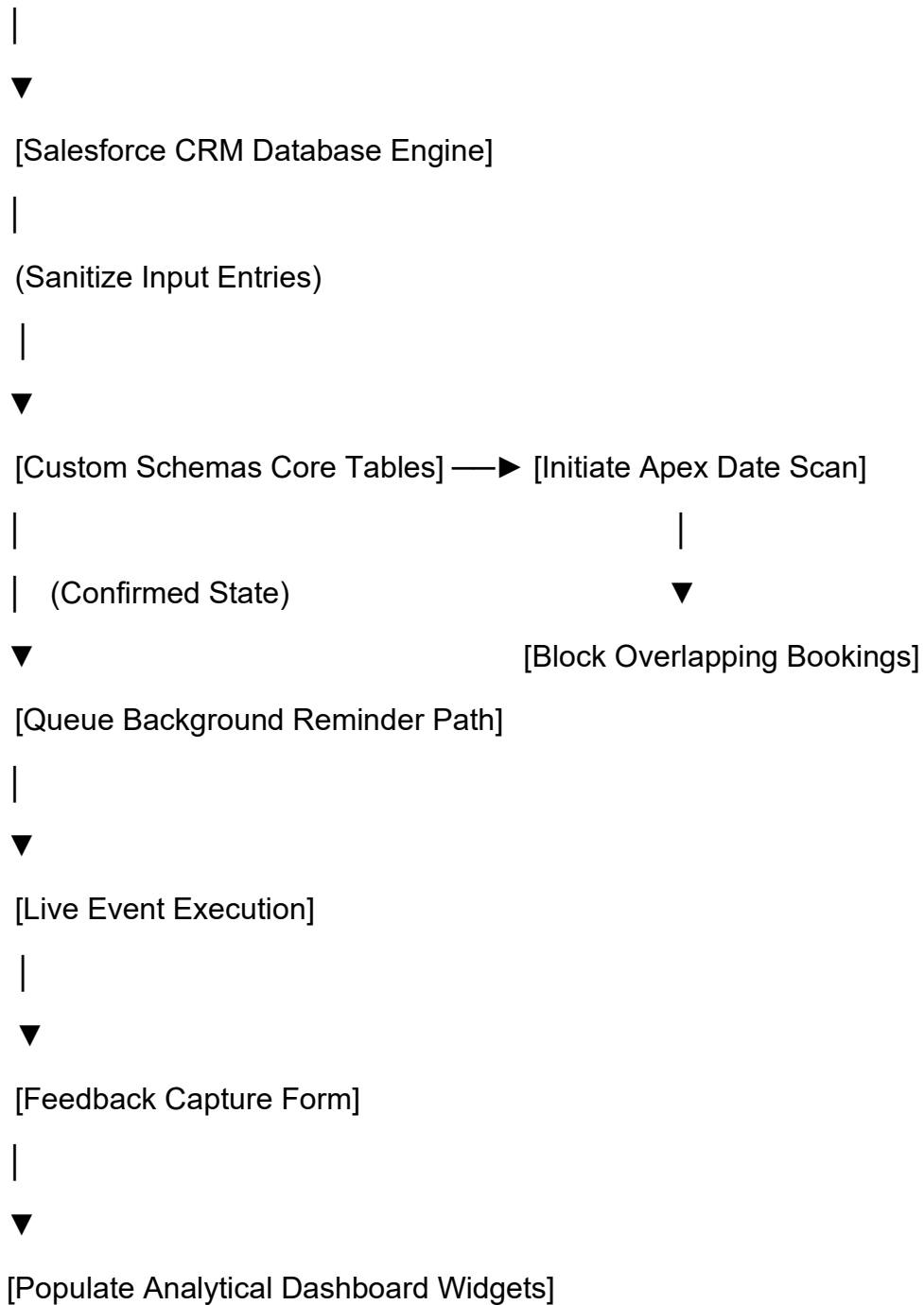
- **FR-10:** Operational intelligence charts (populating graphical widgets from custom summary data).
- **FR-11:** Event status lifecycle automation (monitoring timelines to auto-close expired rows).
- **FR-12:** Unified utility data migration (cleaning legacy customer entries during software ingestion).

3.3 Non-Functional Requirements

- **Usability:** Custom Lightning app application delivers clean layout paths and tab navigation structures.
- **Security:** Enforces strict role-based data boundaries to guarantee explicit row-level record isolation across profiles.
- **Reliability:** Implements data-hygiene validation checks to prevent incorrect database write actions.
- **Performance:** Background process components leverage bulkified logic loops to execute within standard cloud governor parameters.
- **Availability:** High system uptime backed by secure multitenant cloud architecture.
- **Scalability:** Metadata structures accommodate growing transactional data and high concurrent user volume easily.
- **Maintainability:** Uses standard declarative tools where possible, allowing quick rule adjustments without custom programming.
- **Data Integrity:** Implements referential key constraints and context filters to block orphan database rows.

3.4 Data Flow Diagram

[User Interface Intake Node]



Detailed Process Workflows

1. **Cancellation Automation Process:** When a cancellation request is filed, the system switches the status field to *Pending Cancellation* and restricts user edit capabilities. The record routes directly to the assigned Manager's review queue, updating fields automatically based on approval or rejection.
2. **Property Validation Pipeline:** The system scans reservation rows when schedule fields update. The background engine cross-checks the target venue and date parameters against existing rows. If an overlap is flagged, it stops database commitment; otherwise, the space status switches to *Reserved*.
3. **Nightly Lifecycle Maintenance Process:** A scheduled automated process updates rows at 8:00 PM every night. The background engine checks the calendar timeline for past event dates, changing active status markers to *Completed* safely.

3.5 Technology Stack

Infrastructure Layer	Selected Technical Tool	Architectural Purpose inside System
Environment Org	Salesforce Developer Edition	Core database hosting & setup workspace.
Presentation View	Lightning Application Workspace	Single-page browser workspace tab groups.
Schema Storage	Custom Database Tables (___c)	Multi-tenant row-and-column transaction logs.
Declarative Rules	Process Flows & Case Formulas	Automated field budget calculation rules.
Transaction Guard	Standard Approval Engine Channels	Locked stage-change manager review loops.
Backend Engine	Apex Programming (Triggers/Batch)	Run-time conflict prevention checks.
Analytics Engine	Matrix Reports & Dashboard Charts	Visual trend summarization component hubs.

- **Cloud Application Tier:** Salesforce CRM (Lightning Platform Architecture).
- **User Interface Environment:** Salesforce Lightning Experience Framework (Custom Workspace).

- **Data Layer Storage Tier:** Salesforce Custom Metadata Object Framework.
- **Declarative Optimization Tier:** Input Validation Rules, Field Expressions, and Cross-Object Formula Logic.
- **Workflow Coordination Tier:** Record-Triggered Flows and Multi-Tier Approval Process Wizards.
- **Programmatic Core Engine Tier:** Apex Automation Classes, Database Execution Triggers, and Scheduled Batch Routines.
- **Platform Access Controls Tier:** Organization-Wide Defaults, Structural Roles, Sharing Settings, and Custom System Profiles.
- **Business Analytics Tier:** Native Data Reports Engines and Graphical Metrics Dashboard Canvases.

4. PROJECT DESIGN PHASE

Solution Architecture & Object Fields Topology

Object Label	API Structural Name	Salesforce Relationship Type	Custom Data Fields & Picklist Options
Event	Event__c	Master Hub	Event Name (Text), Event Date (Date), Event Type (Picklist: Wedding, Corporate, Birthday, Anniversary, Festival, Concert, Other), Event Status (Picklist: Planned, Confirmed, Completed, Pending Cancellation, Canceled, Rejected), Event Budget (Formula: Currency return type based on Event Type CASE statement).
Client	Client__c	Parent Lookup	Client Name (Text), Email (Email), Phone (Phone), Address (Text Area), Country (Picklist), City (Picklist).
Vendor	Vendor__c	Parent Master-Detail	Vendor Name (Text), Email (Email), Phone (Phone), Service Type (Multi-Select Picklist: Catering, Decor, Photography, Videography, Lighting, Stage Setup, Makeup Artist, DJ/Music, Transportation, Hosting/Anchor), Status (Picklist: Available, Booked, Cancelled).

Venue	Venue__c	<i>Parent Lookup</i>	Venue Name (Text), Address (Text Area), Location (URL), Capacity (Number), Availability Status (Picklist: Available, Reserved, Not Available).
Feedback	Feedback__c	<i>Child Lookup</i>	Feedback ID (Auto-Number: F-{{0000}}), Rating (Picklist: 1, 2, 3, 4, 5), Comments (Long Text Area).
Event Vendor	EventVendor__c	<i>Junction Object</i>	Event__c (Master-Detail link to Event), Vendor__c (Master-Detail link to Vendor).

4.1 Problem–Solution Fit

- **Fragmented spreadsheet logs:** Custom Salesforce Objects → Centralized event data.
- **Manual booking steps:** Salesforce Flows → Reduced manual effort.
- **Venue scheduling overlaps:** Apex Trigger and Venue Status Helper → Prevents booking conflicts.
- **Disconnected partner records:** Event Vendor Junction Object → Supports complex vendor assignment.
- **Delayed client updates:** 3-Day Asynchronous Reminder Flow → Timely communication.
- **Unstructured cancellation steps:** Managed Approval Pipelines → Controlled approval governance.
- **Malformed data inputs:** Input Validation Rules → Clean profile creation.
- **Limited management metrics:** Report Engines and Analytics Dashboards → Better monitoring.

- **Unsecured record exposure:** Custom Profiles and OWD Rules → Controlled secure access.

4.2 Proposed Solution

- **The Operational Problem:** Manual record-keeping via scattered tracking files.
- **The Selected Strategy:** Deploying an integrated cloud database application on the Salesforce CRM infrastructure.
- **Automation Focus:** Setting up background flows for customer reminders, apex triggers for venue protection, and formalized approval workflows for cancellations.
- **Strategic Outcome:** Unified operational transparency, protected client records, and live report summaries for corporate executives.
- **Platform Growth:** Engineered on metadata boundaries to handle future record increases and team growth smoothly.

4.3 Solution Architecture

The operational framework connects Custom Objects, Relationship Fields, Validation Rules, Record-Triggered Flows, Managed Approval Pipelines, Apex Classes, Sharing Settings, Custom Profiles, and Analytics Dashboards into a scalable data environment.

5. PROJECT PLANNING AND SCHEDULING

5.1 Methodology

The platform development followed an Agile delivery methodology. Core requirements were broken down into individual user stories, added to a product backlog backlog, and given point weights using abstract estimation to track velocity across sprints.

5.2 Product Backlog

Target Sprint	User Story Objective	Assigned Story Points	Core Implementation Priority
Sprint 1	Create type, date, and status logs.	3 Points	High Priority
Sprint 1	Map clients and venues to events.	5 Points	High Priority
Sprint 2	Link multiple suppliers to an event.	5 Points	High Priority
Sprint 2	Execute timely alert notifications.	3 Points	High Priority
Sprint 3	Secure a controlled manager workflow.	5 Points	High Priority
Sprint 3	Prevent venue scheduling double-bookings.	8 Points	High Priority
Sprint 4	Lock role hierarchy access levels.	5 Points	High Priority
Sprint 4	Deploy tracking metrics dashboards.	5 Points	Medium Priority

- **Sprint 1 (Event Core Tracking Module):** User Story: As an Event Coordinator, I want to create events with type, date and status. Points: 3. Priority: High.
- **Sprint 1 (Client & Venue Referential Links):** User Story: As an Event Coordinator, I want to associate a client and venue with an event. Points: 5. Priority: High.
- **Sprint 2 (Logistics Junction Assignment Tier):** User Story: As a Vendor Manager, I want to assign multiple vendors to an event. Points: 5. Priority: High.
- **Sprint 2 (Asynchronous Notification Inbound):** User Story: As a Client, I want timely event reminders. Points: 3. Priority: High.
- **Sprint 3 (Managed Cancellation Governance):** User Story: As an Event Coordinator, I want cancellation requests to follow an approval process. Points: 5. Priority: High.

- **Sprint 3 (Programmatic Collision Prevention Trigger):** User Story: As a Venue Manager, I want double bookings prevented. Points: 8. Priority: High.
- **Sprint 4 (Least-Privilege Security Access Rules):** User Story: As an Event Administrator, I want role-based access. Points: 5. *Priority: High.*
- **Sprint 4 (Executive Metrics Intelligence Console):** User Story: As Management, I want reports and dashboards. Points: 5. Priority: Medium.

6. PROJECT DEVELOPMENT PHASE

PHASE 1: REQUIREMENT ANALYSIS & PLANNING

The initial phase converted manual business operations into explicit system parameters. Developers designed metadata object structures, relationship mappings, validation metrics, and profile security tiers to create a strong platform design baseline before configuration.

PHASE 2: BACKEND DEVELOPMENT & CONFIGURATION

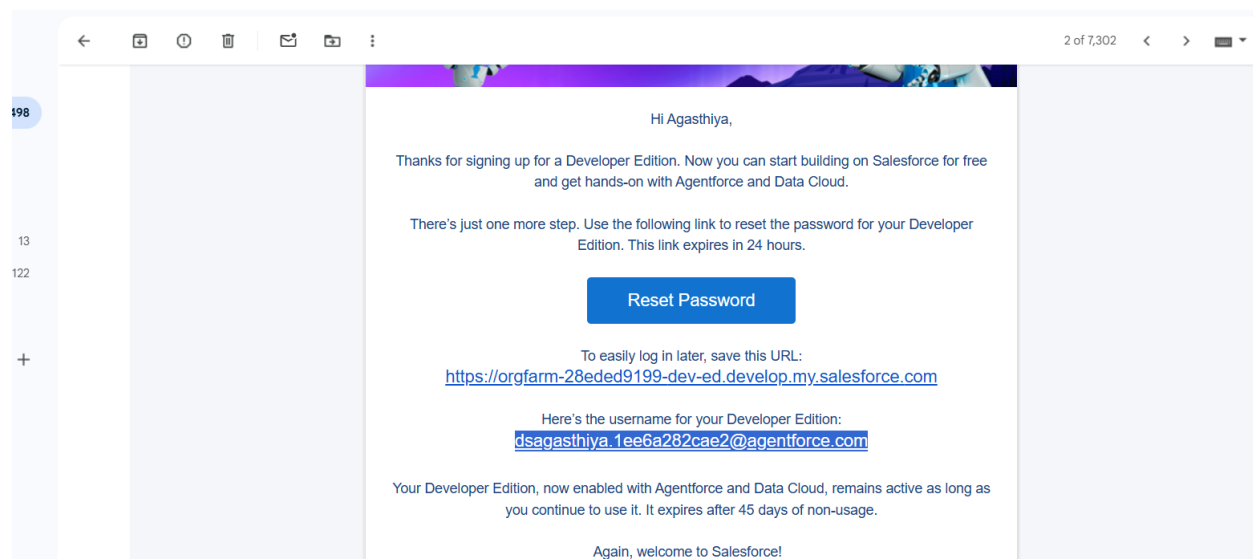
The core setup phase configures metadata fields and builds out relationships across custom tables. Declarative tools manage user input filters, validation check blocks, and approval tracking states, while custom programming supports advanced scheduling validation. Access control configurations leverage sharing settings and profile blocks to ensure client data protection across operations.

MILESTONE 1 – SALESFORCE ACCOUNT

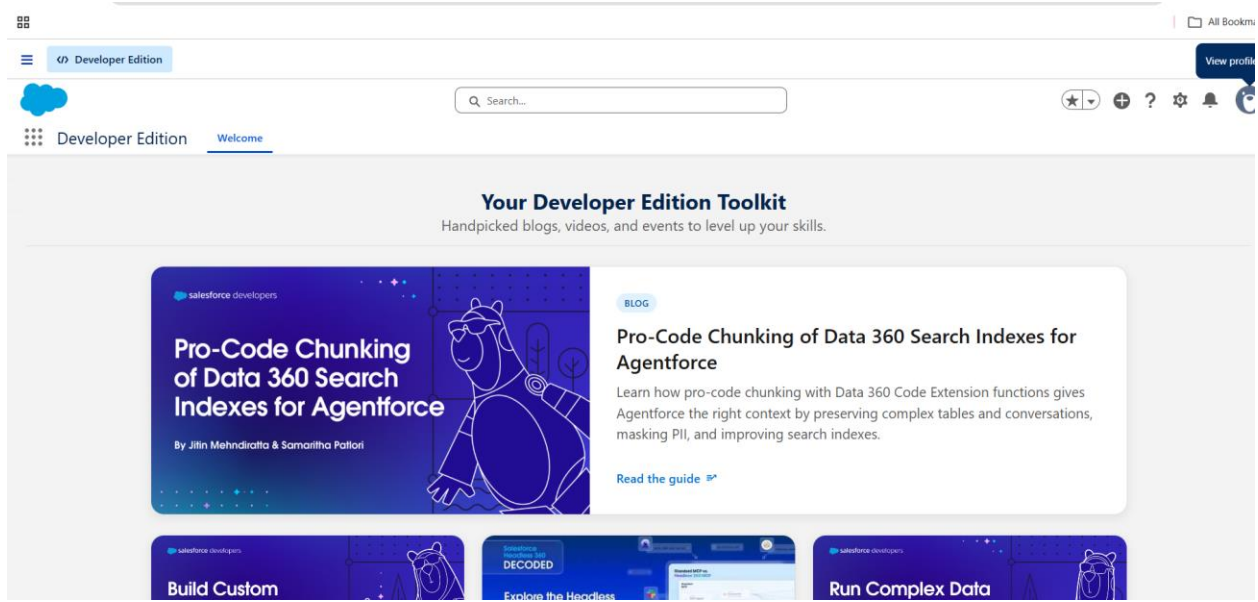
Salesforce is a cloud-based CRM platform used to manage customer relationships, business data, automation, collaboration, reporting, and analytics. Standard cloud components store data securely in the cloud while allowing users to manage relationships with prospects and customers.

The system configuration menu handles metadata adjustments and sets up background tracking rules across our backend tables.

Upon successful verification of the platform environment credentials, a system generated activation notification is dispatched to finalize the system password enrollment policies.



The newly provisioned developer environment routes directly to the platform toolkit console, verifying successful instance registration.

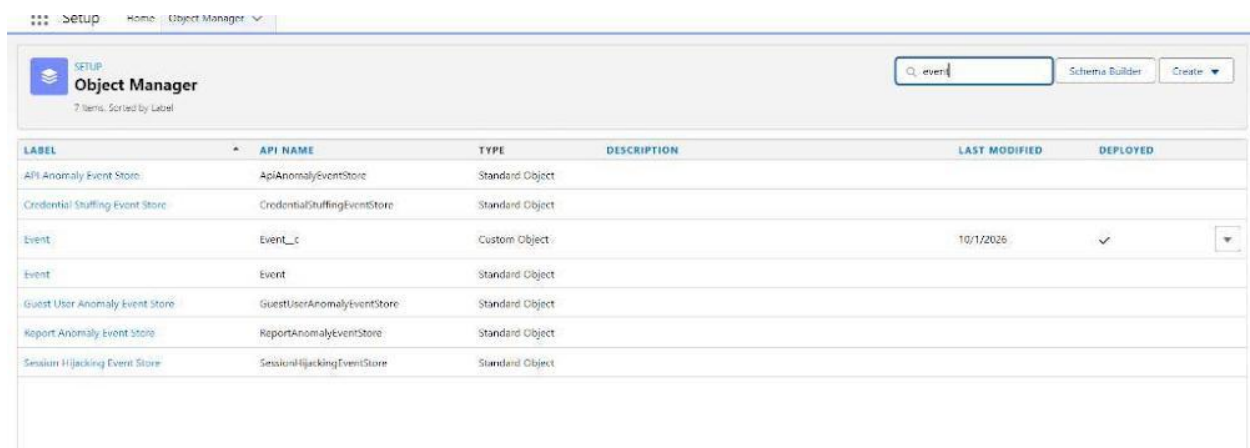


MILESTONE 2 – CUSTOM OBJECTS

Custom objects function as relational database tables, creating unique fields and rows to structure event specifications cleanly.

The metadata engine controls historical log recording and activates backend tracking tools across custom object schemas.

The centralized system object manager workspace indexes custom metadata properties, confirming successful registration of the main application tables within the schema:



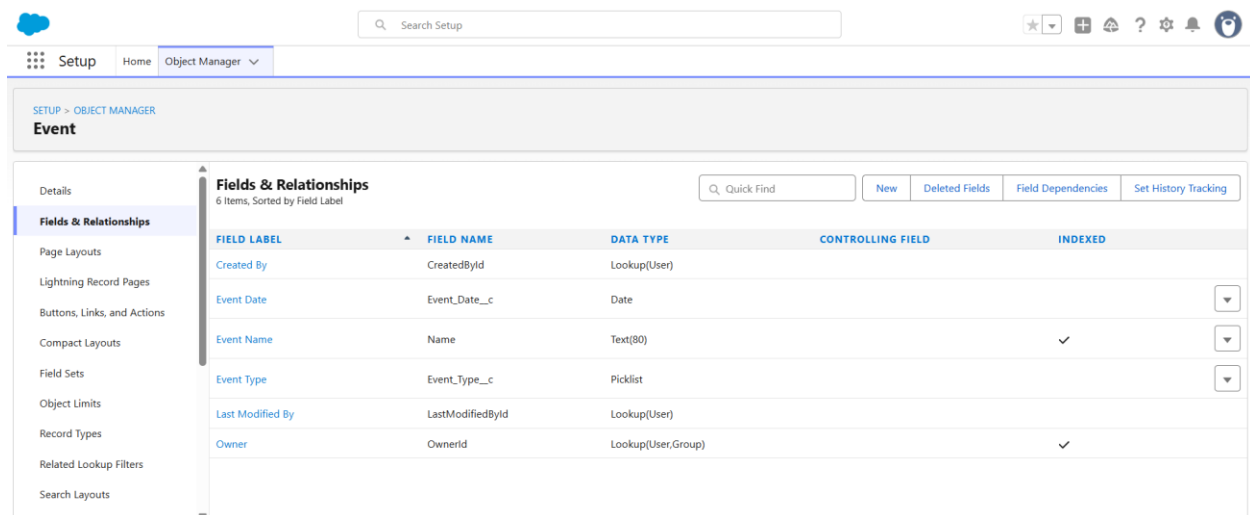
Object Manager

7 Items, Sorted by Label

Label	API Name	Type	Description	Last Modified	Deployed
API Anomaly Event Store	ApiAnomalyEventStore	Standard Object			
Credential Shuffling Event Store	CredentialShufflingEventStore	Standard Object			
Event	Event__c	Custom Object		10/1/2026	✓
Event	Event	Standard Object			
Guest User Anomaly Event Store	GuestUserAnomalyEventStore	Standard Object			
Report Anomaly Event Store	ReportAnomalyEventStore	Standard Object			
Session Hijacking Event Store	SessionHijackingEventStore	Standard Object			

The system realization deploys six explicit custom tables to organize operations:

Object Label	API Structural Name	Core Database Role
Event	Event__c	Houses primary event transactional entries.
Client	Client__c	Maintains the master customer index.
Vendor	Vendor__c	Indexes logistical contract profiles.
Venue	Venue__c	Tracks facility maps, limits, and spaces.
Feedback	Feedback__c	Captures customer reviews and ratings.
Event Vendor	EventVendor__c	Junction table fixing many-to-many links.

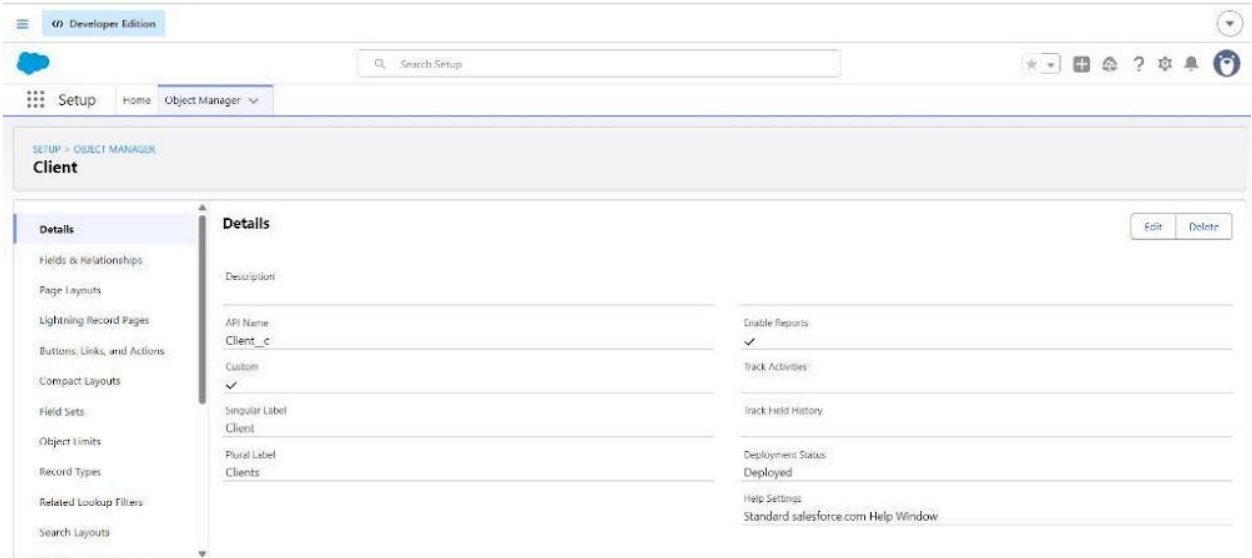


Event

6 Items, Sorted by Field Label

Field Label	Field Name	Data Type	Controlling Field	Indexed
Created By	CreatedById	Lookup(User)		
Event Date	Event_Date__c	Date		
Event Name	Name	Text(80)		✓
Event Type	Event_Type__c	Picklist		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓

- Event Entity (Event__c):** Houses primary records including dates, type tags, execution statuses, and automated budget metrics.
- Client Entity (Client__c):** Maintains the master customer index, tracking phone contacts, verified email strings, and street addresses.



3. **Vendor Entity (Vendor__c):** Indexes external logistical contractors, including field classifications, communication files, and contract flags.
4. **Venue Entity (Venue__c):** Tracks facility locations, max space limits, physical maps links, and active reservation parameters.
5. **Feedback Entity (Feedback__c):** Captures customer reviews, descriptive comments rows, and numerical rating metrics.
6. **Event Vendor Junction Object (EventVendor__c):** Connects vendor rows to separate events via dual Master-Detail relationships, resolving the native many-to-many link.

MILESTONE 3 – TABS

Custom object tabs provide user-friendly interface access points inside the navigation menu. The application provisions specialized structural tabs for Events, Clients, Vendors, Venues, and Feedbacks to ensure fast daily access.

These tabs are integrated into the custom Event Planner Lightning Application utility bar to allow multi-tab navigation, enabling coordinators to manage client profiles and event calendars simultaneously without losing active session data.

MILESTONE 4 – FIELDS AND RELATIONSHIPS

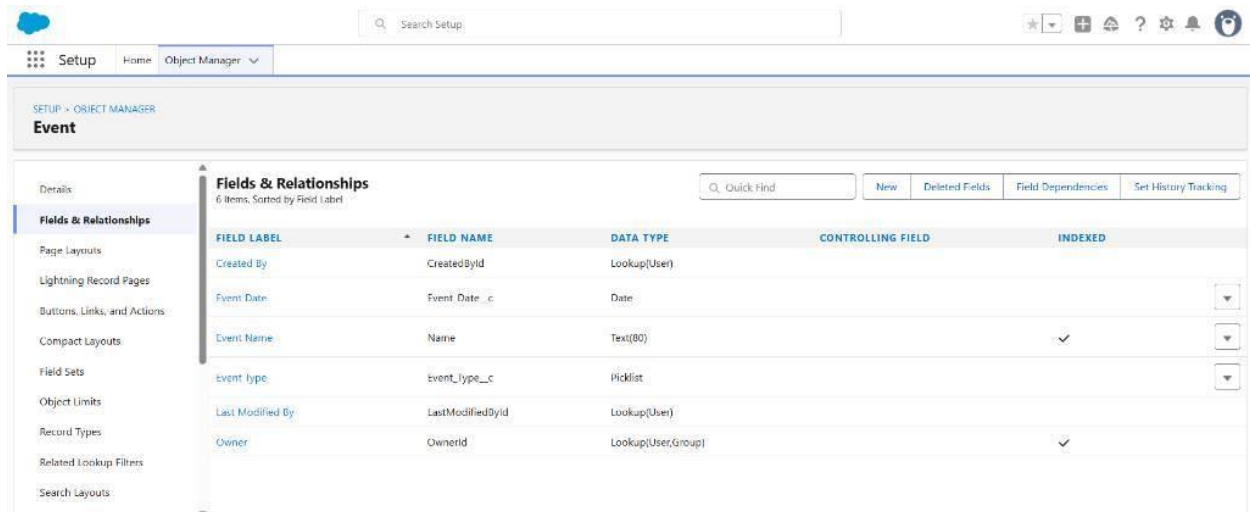
Custom fields match database columns, storing parameters required to structure corporate transactions safely across tables.

Event Object Field Definitions

Field Label	API Structural Name	Data Type / Picklist Options
Event Type	Event_Type__c	Picklist: Wedding, Corporate, Birthday, Festival, etc.
Event Date	Event_Date__c	Date field format.
Event Status	Event_Status__c	Picklist: Planned, Confirmed, Completed, Cancelled.
Event Budget	Event_Budget__c	Formula (Currency return type via CASE logic).

- **Event Type Picklist (Event_Type__c):** Wedding, Corporate, Birthday, Anniversary, Festival, Concert, Other.
- **Event Date Field (Event_Date__c):** Standard Date data type configuration.
- **Event Status Picklist (Event_Status__c):** Planned, Confirmed, Completed, Pending Cancellation, Cancelled, Rejected.
- **Event Budget Formula Field (Event_Budget__c):** Automated currency field that evaluates picklist parameters to output standardized budget metrics.

The custom fields manager interface updates dynamically to track newly engineered column attributes within the object schema definitions:



Relational References Infrastructure

- **Lookup Fields Architecture:** Employs standard Lookup Relationships to connect the Event master record to Client profiles and Venue assets safely.
- **Junction Table Master-Detail Rules:** Links EventVendor__c records directly to parent objects via Master-Detail fields. If a parent event file is purged, the database automatically deletes all dependent junction cards.
- **Feedback Referral Pathways:** Lookup fields map user review lines directly back to specific events and customer accounts.

Contextual Lookup Filter Controls

To eliminate user data entry errors during feedback logging, an active Lookup Filter is deployed on the Event lookup field inside the Feedback object.

- **Filter Criteria Enforced:** Feedback: Client__c EQUALS Event: Client__c.

Filter Property	Configuration Definition	Enforced Runtime Behavior
Source Field	Feedback__c.Event__c	Restricts searchable records on form load.
Matching Rule	Feedback Client == Event Client	Filters out unlinked historical records.
System Goal	Data Entry Governance	Stops coordinators matching wrong events.

- **Behavior:** The interface dynamically filters out unlinked records, displaying only event files matching the selected customer profile.

MILESTONE 5 – VALIDATION RULES

Validation rules verify that character values meet system data quality rules before records can save to the cloud storage layers.

- **Rule Unique Name:** Email_Valid_Address

Metadata Property	System Configuration Value
Rule Name	Email_Valid_Address
Formula Logic	NOT(REGEX(Email__c, "^[a-zA-Z0-9._]+@[a-zA-Z0-9._]+\.[a-zA-Z]{2,}\\$"))
Error Message	"Please Enter Valid Email Address"
Error Location	Field-level (Email)

- **Expression Script Function:** Evaluates client email fields against regular expression parameters. Text patterns that fail format parsing throw the system layout error message "Please Enter Valid Email Address" and halt saving.

MILESTONE 6 – APPROVAL PROCESS

The cancellation process handles event status updates and alerts owners systematically when changes are submitted.

- **Approval Reference Key:** Event_Cancellation_Process

Workflow Step	Evaluated System Action	Enforced Database State
Entry Criteria	Event Status updates to Pending Cancellation.	Locks record against user editing.
Initial Action	Fires template notifications to coordinator roles.	Enters manager dashboard review loop.
On Approval	Field Update changes status to Cancelled.	Dispatches confirmation email alert to client.
On Rejection	Field Update resets status back to Confirmed.	Reverts records to active calendar state.

- **Process Parameter Constraints:** Triggers when an active status updates to Pending Cancellation. It locks records against user editing and routes alerts to the defined approver role, changing statuses dynamically based on final review.

7. AUTOMATION ENGINE & SYSTEM ANALYSIS

- **Time-Dependent Architecture:** Configured an action-optimized scheduled path within the core Salesforce workflow engine that continuously references the Event_Date__c field to automatically evaluate, track, and dispatch standard transactional reminder notifications to clients exactly **3 days prior to execution** with automated owner carbon-copy (CC) routing.

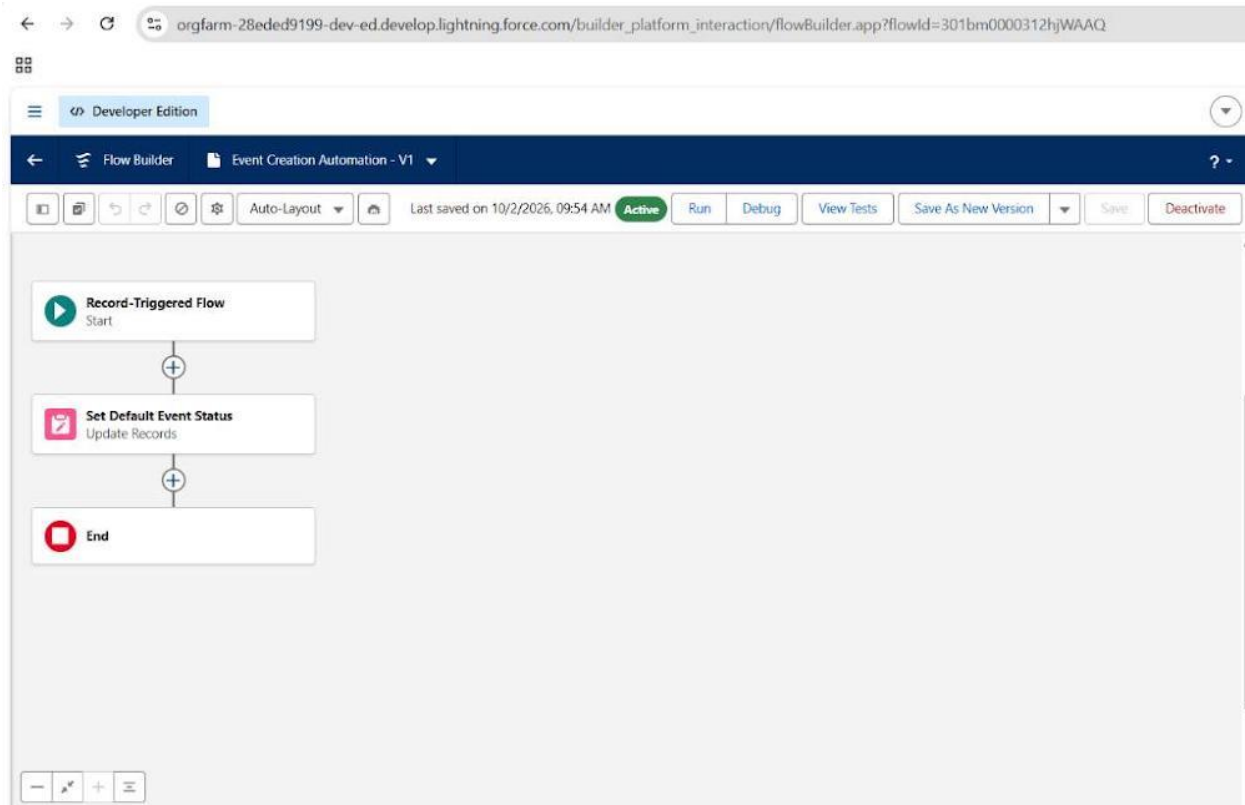
- **Advanced Time-Dependent Scheduling Pathways:** The background engine operates using a scheduled path mechanism that tracks the target calendar date and automatically runs notification loops exactly three days before the event, eliminating manual monitoring tasks for event coordinators.
- **Asynchronous Resource Allocation Management:** The flow is optimized for Actions and Related Records, allowing the system to fetch cross-object client data fields asynchronously without impacting the performance of live database transactions or hitting platform execution limits.

3-Day Event Reminder Flow

Salesforce flows automate time-dependent processes and send client notifications based on criteria rules without requiring manual execution.

- **Central Platform Management:** Acts as a **single data hub** where users can easily track events, clients, vendors, and venues in one place instead of using messy spreadsheets.
- **Automates Tasks:** Reduces manual workload by automatically running time-based workflows like email reminders and cancellation approvals.

The record-triggered flow engine tracks real-time data adjustments and runs automated background tasks when event records are initialized within the ecosystem:



- **Trigger Node Evaluation:** Monitors the Event object when records are created or updated to meet entry filters: `Event_Status__c = Confirmed`.
- **Optimization Layer Mode:** Optimized to run background actions and handle cross-object relationship variables smoothly.

Action Module Search & Extraction Routing

When an event record enters this path, the system opens its action element library to locate the email sending module.

The automation framework extracts field values from the triggering row using system variables.

The recipient field is dynamically mapped to the client email address field on the related customer record, ensuring accurate message delivery.

- **Scheduled Path Offset Parameters Matrix:**

- Path Developer Key: 3_Days_Before
- Time Source Field: Event__c.Event_Date__c
- Offset Number Sizing: 3
- Interval Selection Type: **Days Before**

Venue Double-Booking Prevention

An automated check runs before database write operations to scan scheduled event timelines and prevent duplicate venue bookings on matching dates. If an overlap is flagged across records, the engine blocks the save operation.

Event Completion Automation

Scheduled background tasks query old entries at night, checking for passed calendar dates to update matching event rows to Completed status automatically.

8. USER INTERFACE & REPORTING

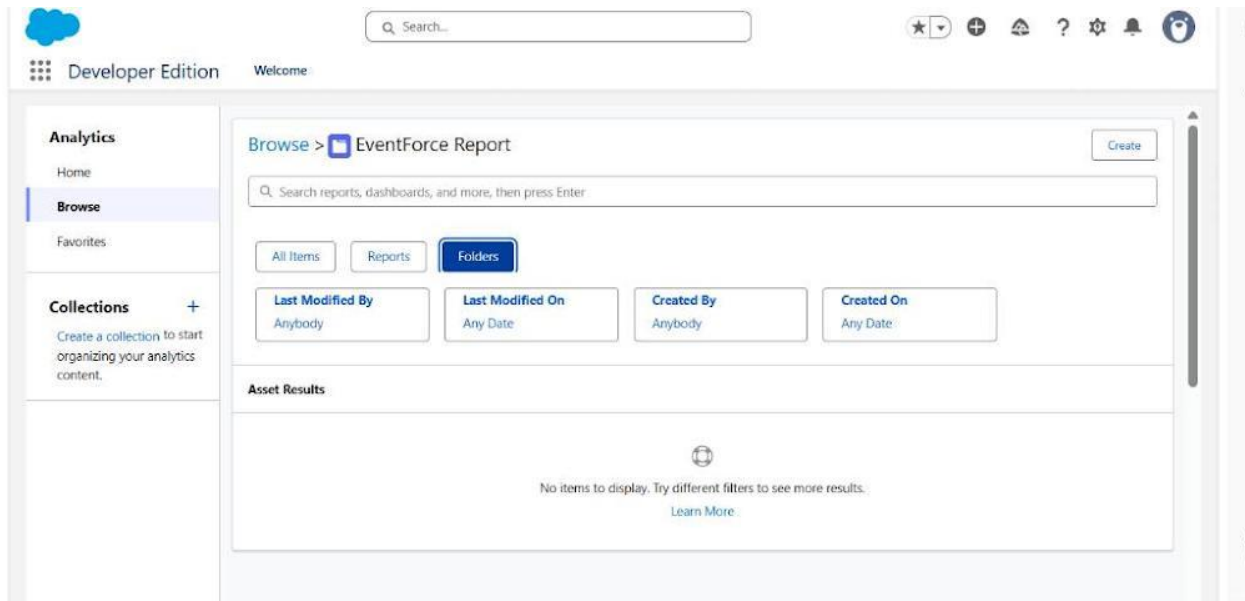
PHASE 3 – USER INTERFACE & REPORTING

The custom Lightning Application delivers a unified operations workspace, providing structured navigation tabs for Events, Clients, Vendors, Venues, Feedback entries, and custom Dashboard consoles.

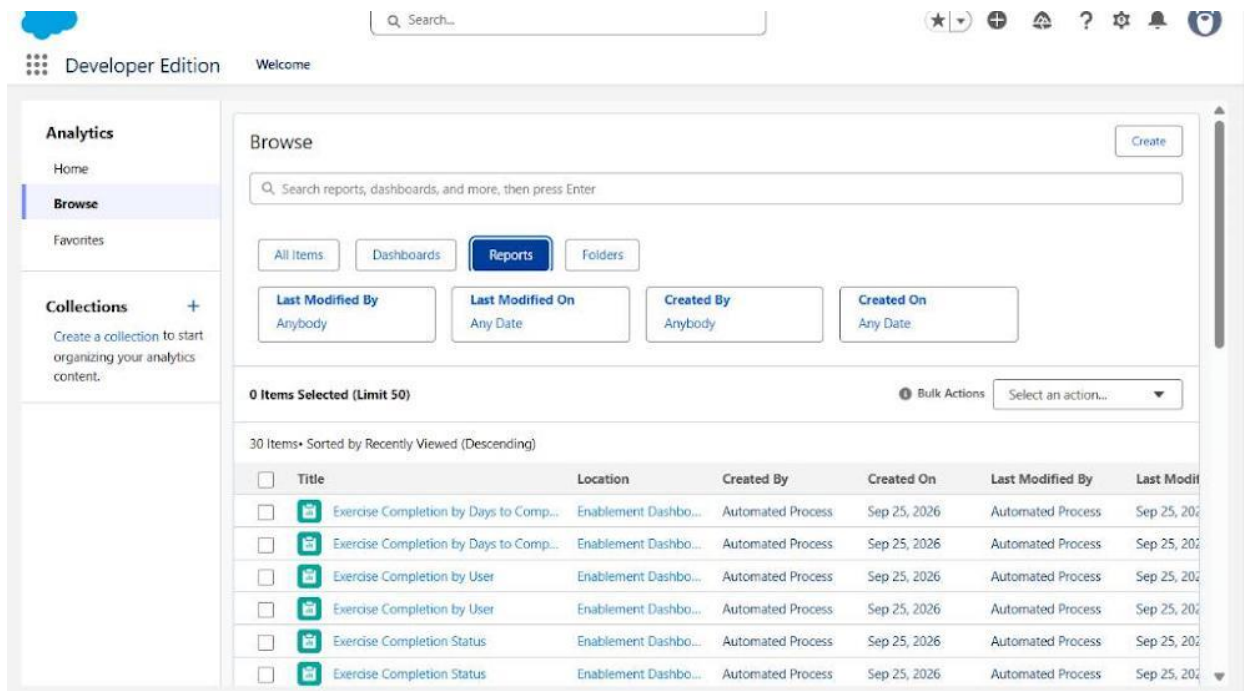
REPORTS

The system organizes transactional records into standard matrix templates to track operational metrics over time.

The reporting engine groups entries to filter out canceled bookings and show total business numbers by calendar months.



- **Organizes Grouped Data:** Groups active records dynamically by **Calendar Month** to let managers spot busy seasonal trends instantly.
- **Shows Total Budget:** Automatically calculates the **Sum of Event Budget** metrics so leadership can monitor overall monthly company revenue.
- **Removes Canceled Entries:** Uses built-in filters (Event Status Not Equal To Cancelled) to ensure only actual, confirmed upcoming events are included in your analysis.



Pipeline Report Summary Framework

- **Report Name Designation:** Upcoming Events by Month
- **Data Fields Included:** Event Name, Event Date, Event Type, Client Portfolio, Venue Name, calculated Event Budget.
- **Active Logic Filters:** Show Me = All Events | Event Date $\geq$ TODAY | Event Status $\neq$ Canceled.
- **Monthly Trend Chart:** Displays the data as a clean **Column or Line chart** to let managers see how many events are booked each month at a glance.
- **Auto-Updated Data:** Uses the platform's live reporting engine to **automatically refresh** components as soon as individual event details change.
- **Shared Folder Access:** Saved securely inside the **EventForce Reports** folder with viewing permissions restricted to team roles like coordinators and managers.

New Events Report

Events

Outline

Filters 1

Filters

Add filter...

Show Me
All events

Event Date
Today (On Oct 1, 2026)

Event Date

Previewing a limited number of records. Run the report to see all records.

Event Date	Event: Event Name	Event Type	CI

Filter By

Field

Event Date

Operator

equals

Type

Value

Date

Use relative date

Pick a date

☐ Locked

Cancel

Apply

- **Grouping Setup:** Grouped Rows by Event Date (Segmented by Calendar Months).
- **Live Report Customization:** The active report creation window allows developers to add direct filters, edit layout criteria, and adjust row mappings on the fly.
- **Date Filter Rule:** Uses a dynamic date filter set to Event Date greater than or equal to TODAY to make sure the report only shows upcoming bookings.
- **Row Count Summary:** Automatically calculates the total number of event entries within each month to provide an exact snapshot of workload distribution

Add Chart

Save & Run

✔

Previewing a limited number of records. Run the report to see everything.

Event Date ↑ ▾

Event: Event Name ▾

Event Type ▾

Client ▾

Venue ▾

Event Budget ▾

No records returned in preview. Try running the report or editing report filters.

• Set the Event Date filter to [All Time](#).

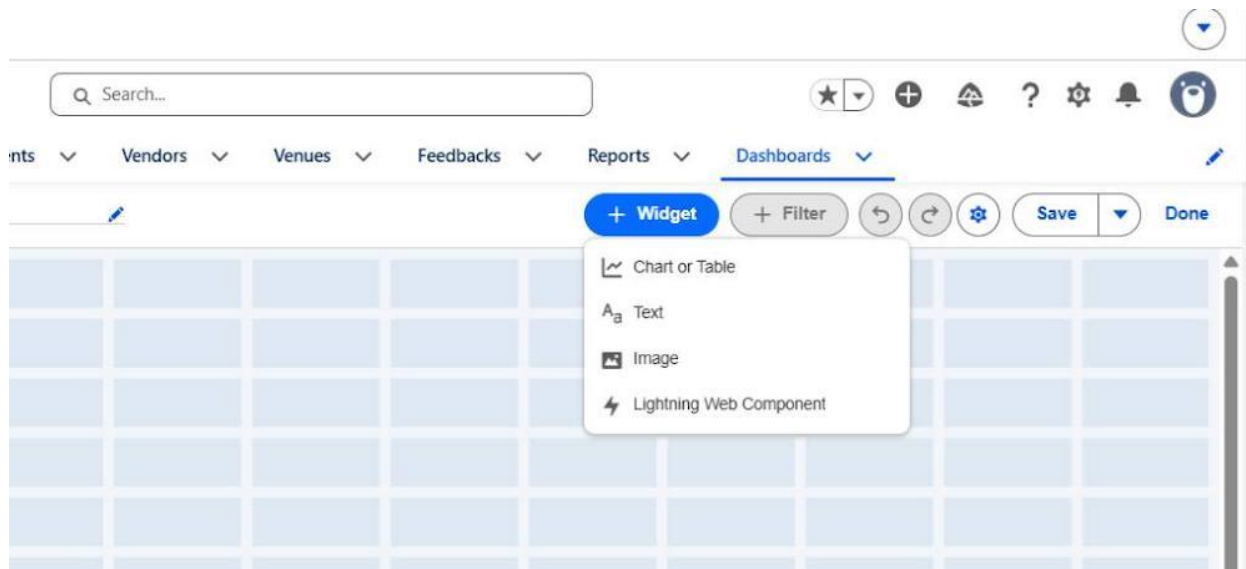
• Edit other filters in the filter panel.

DASHBOARD

The dashboard architecture combines summary data tables into high-level visual charts, giving managers an instant look at pipeline operations.

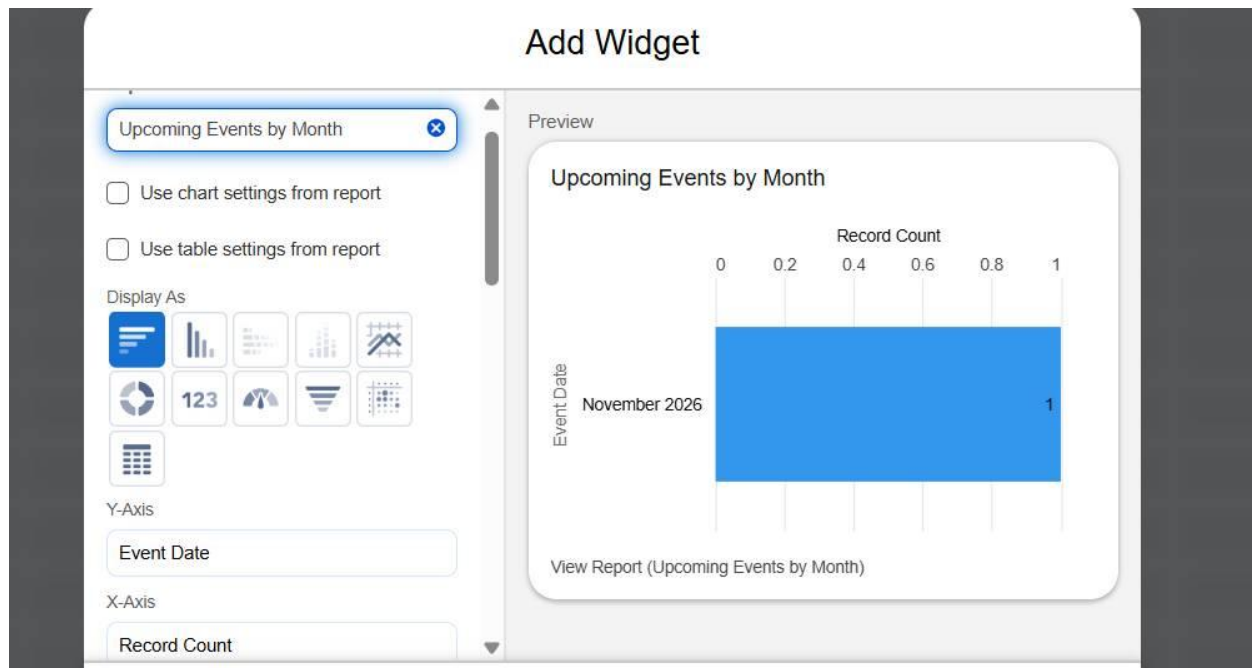
- **Real-Time Key Performance Indicator Aggregation:** The console converts complex summary report data into dynamic, high-level chart widgets—such as donut diagrams and grouped metrics bars—enabling executive management to assess event pipeline volume, operational milestones, and budget performance across various departments instantly.
- **Centralised Multi-Metric Operational Intelligence:** The visual layout is built using a custom canvas layout grid that links directly to the underlying data schema, providing users with context-aware data drill-down features to click through chart visuals and inspect source records without opening separate lookup menus.

The interactive dashboard canvas layer provisions visual components utilizing dynamic widget components layout panels to build responsive performance reports summaries:



- **Interactive Design Canvas Layout Grid:** The workspace features an alignment layout canvas background grid that enables administrators to drag, drop, resize, and align separate metric tracking visual modules into a clean dashboard console layout.
- **Dynamic Component Addition Selection Menu:** The active **+ Widget** dropdown feature options menu displays the primary integration layout nodes, allowing designers to select between Chart or Table, Text descriptive boxes, External Images, and custom Lightning Web Components.
- **Contextual Operational Filter Customization Node:** The adjacent **+ Filter** properties selector command interface tool button allows configurations engineers to build dynamic drop-down sorting parameters, enabling operators to filter the live dashboard metrics dynamically by dates, event types, or locations.
- **Global Execution Administration Control Strip:** The operational top toolbar panel houses versioning and status layout management buttons—including Undo/Redo arrows, Dashboard settings configurations icon, Save data button, and a Done execution button—to deploy newly modified dashboard metrics configurations securely.

The component property wizard maps structural source reports to selected visual formatting options to configure chart layouts:

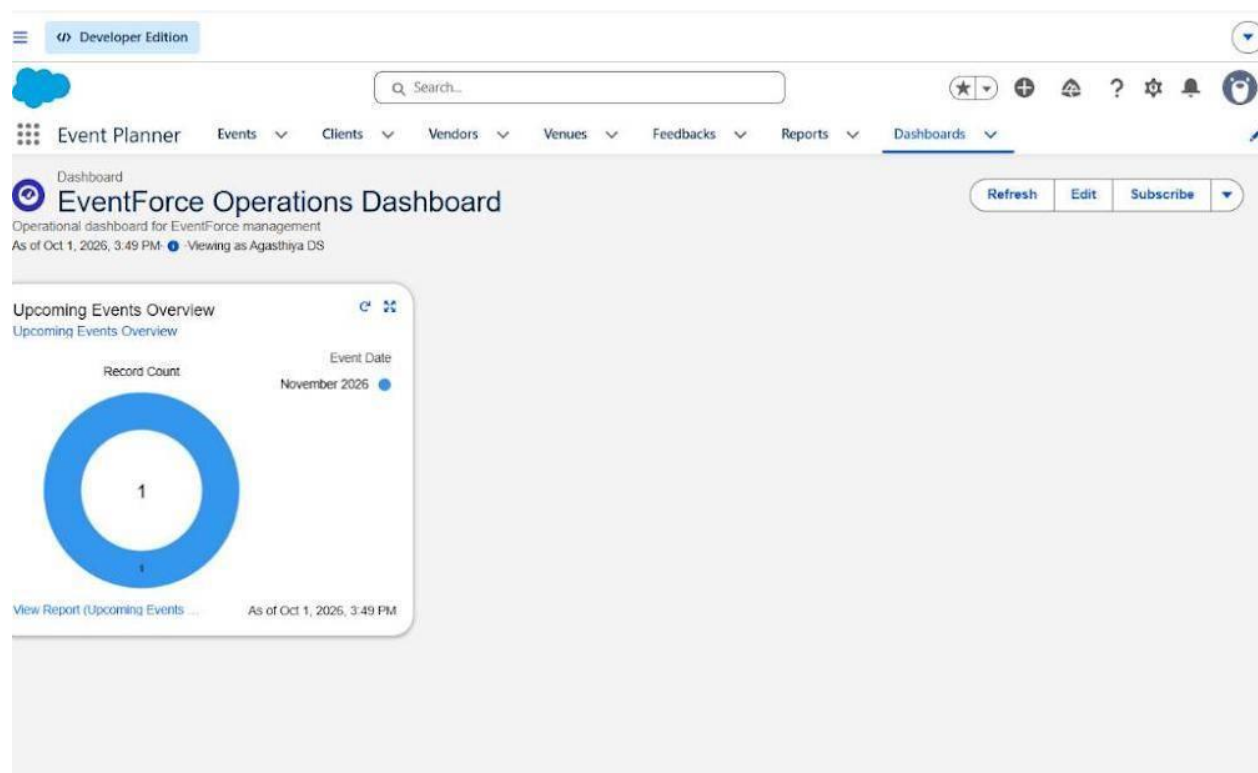


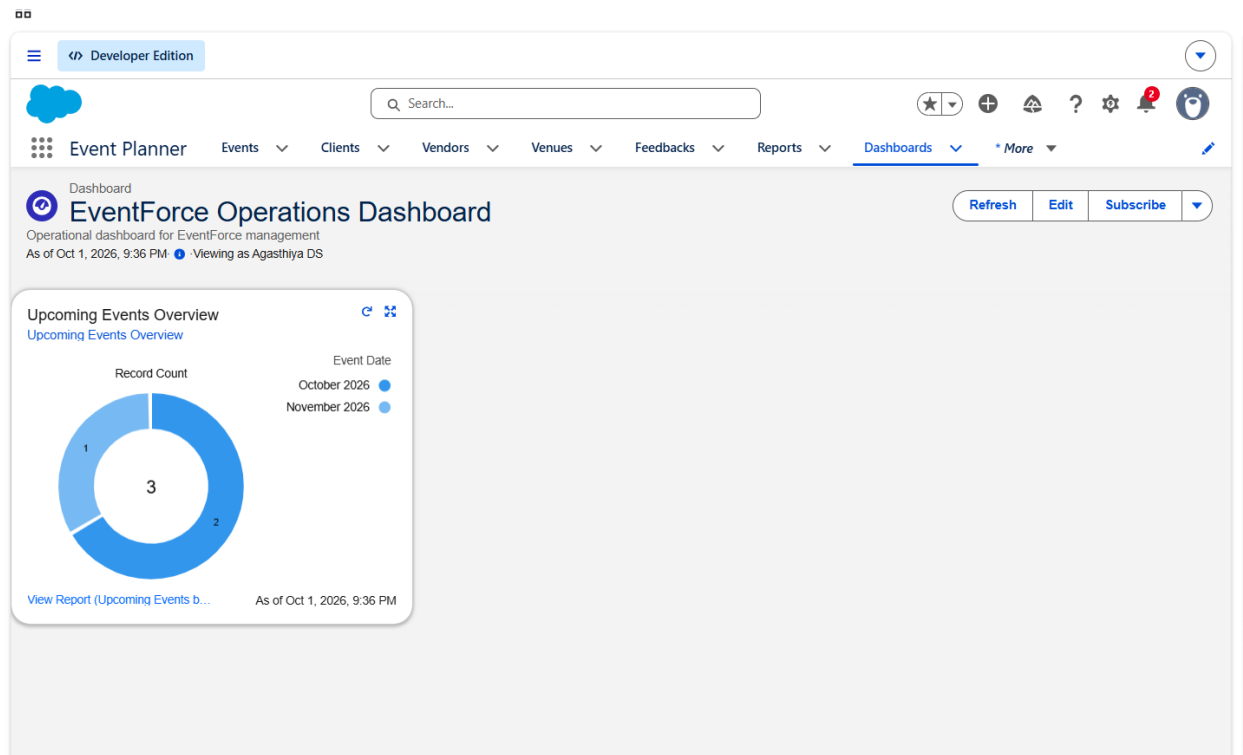
- **Component Property Wizard:** The interface displays an "Add Widget" screen, which serves as a component property wizard designed to map structural source reports to visual formatting layouts.
- **Source Report:** The widget is currently linked to a source report titled "Upcoming Events by Month."
- **Layout and Preview:**
 - The **left panel** contains configuration controls, including display types, axis mapping, and report settings toggles.
 - The **right panel** shows a live preview of a horizontal bar chart displaying a single event record for **November 2026**.
- **Display Configuration:** It offers multiple visual display icons (such as horizontal bar chart, vertical column chart, gauge chart, metric number, and funnel chart) to quickly alter the presentation style.

The completed **Event Force Operations Dashboard** converts data tables into charts, including donut chart components, to display real-time business metrics clearly.

The live production execution dashboard renders a high-level visual console, equipping executives with immediate business metrics intelligence tracking across current pipeline entries:

- **Platform and Dashboard:** The image displays a **Salesforce Developer Edition** platform showing a custom dashboard named "**EventForce Operations Dashboard**," which is used for operational event management tracking.
- **Widget Visualization:** The dashboard contains an "**Upcoming Events Overview**" widget formatted as a blue donut chart, which visualizes a single event scheduled for **November 2026**.





9. DATA MIGRATION, TESTING & SECURITY

SECURITY ARCHITECTURE PARAMETERS

Profile Name	Assigned User License	Object Access Boundaries (CRUD Permissions)	Tab / App Default Workspace Rights
System Administrator	Salesforce	Full CRED (Create, Read, Edit, Delete, View All, Modify All).	Full default visibility across all tabs and administrative tools.
Event Admin	Salesforce	Full CRED (Cloned from System Administrator for total operations management).	Custom default access centered inside the Event Planner App.
Event Coordinator	Salesforce Platform	CRED on Events; Read/Edit on Clients; Read-Only on Venues and Vendors.	Default landing page navigation within the Event Planner App.
Vendor Manager	Salesforce Platform	CRED on Vendors; Read-Only on Events, Venues, Clients, and Feedback.	Visibility restricted to resource planning and matching objects.
Client Profile	Standard Platform User	Read-Only on personal assigned Events; Create/Read on Feedback; No Access on Venues and Vendors.	Access completely restricted to feedback tracking and personal records.

PHASE 4 – DATA MIGRATION, TESTING & SECURITY

Legacy Data Migration Procedures

Data preparation steps clean legacy Excel or spreadsheet entries before importing data rows via the Data Import Wizard utility. The upload preparation verifies formatting parameters, ensures phone data and email fields match text validation patterns, and checks relational links to avoid database errors.

SECURITY ARCHITECTURE PARAMETERS

The system configuration uses strict access filters to protect customer profiles and ensure data isolation across departments.

- **Organization-Wide Defaults (OWD):** Baseline visibility for the Event custom object is set to **Private** for both internal and external access paths, blocking unverified horizontal record exposure across teams.
- **Criteria-Based Sharing Rules:** A custom sharing rule named `Event_Sharing_For_Vendors` handles access extensions. It automatically shares records owned by the Event Coordinator role with users in the Vendor Manager role with Read-Only visibility, supporting business operations safely.
- **Profile Boundaries Settings:** Custom profiles define explicit object-level and field-level permissions, matching platform user actions to job requirements:
 - Event Admin Profile: Cloned from the standard System Administrator base profile. Inherits full data modification rights and programmatic override permissions across objects.
 - Event Coordinator Profile: Built on a platform user license. Grants complete create, read, edit, and delete permissions for events, with

read/edit access for client profiles and read-only views for vendor catalogs.

- Vendor Manager Profile: Cloned from the Standard Platform User profile. It gives full CRED access on Vendors, while restricting Client, Event, Venue, and Feedback objects to Read-Only.
- Client Profile: Built on a platform user license. Restricts users to read-only access for personal events and write access for logging feedback, while unchecking all vendor and venue access boxes.

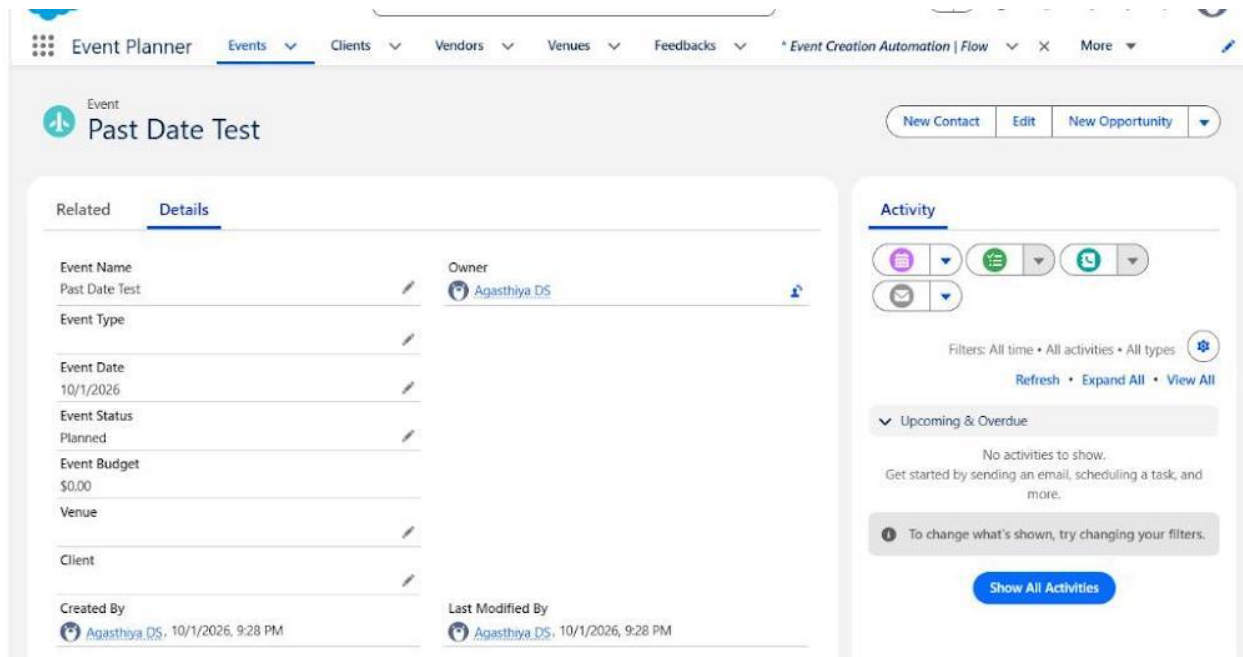
10. SYSTEM MAINTENANCE & TESTING GUIDELINES

FUNCTIONAL & PERFORMANCE TESTING

System verification checks all configurations, automation components, security settings, and dashboard layouts within the developer sandbox instance to confirm operational functionality before deployment.

- **Functional Testing:** Verifies event creation, validation paths, lookup filters, and field formula logic.

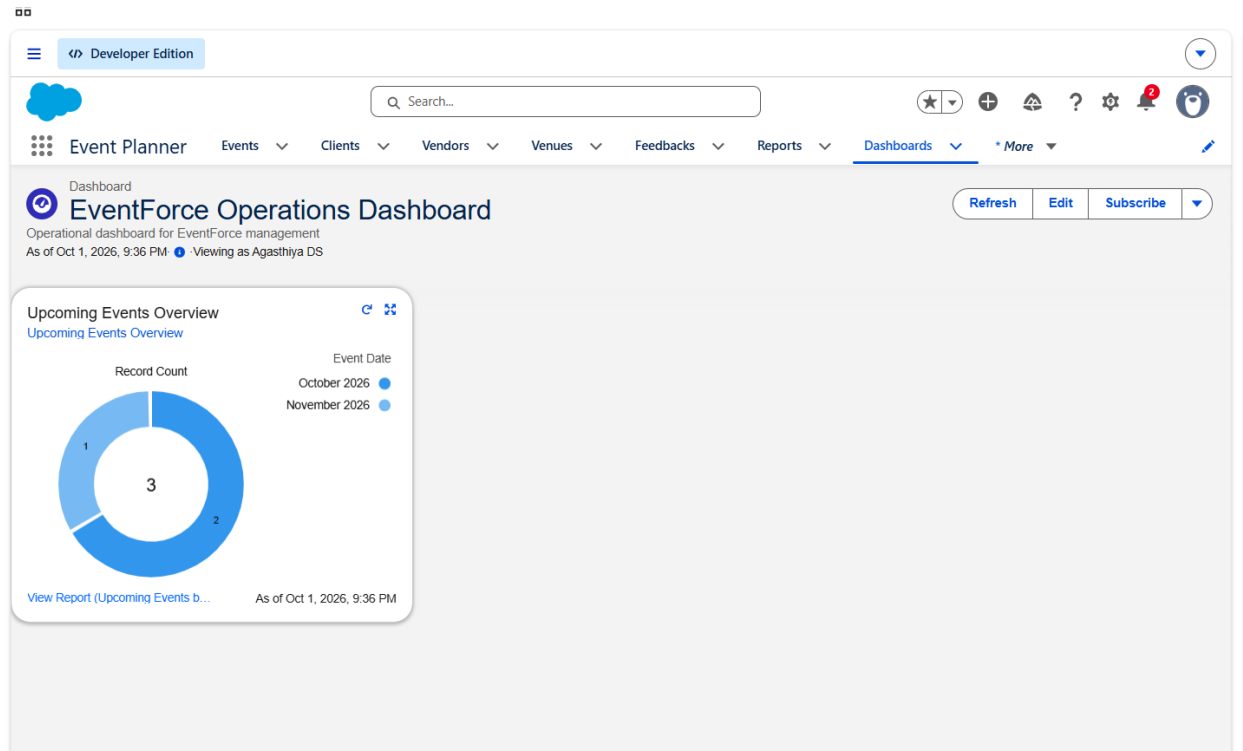
The application runtime validation interface demonstrates a successful functional execution trace when registering a sample testing record within the live production layout:



This image shows a Salesforce event record named "Past Date Test" scheduled for October 1, 2026.

- **Event Status:** It is currently marked as **Planned** with a budget of **\$0.00**.
- **Activity History:** The right panel shows **no active or overdue tasks** logged for this event yet.
- **Dashboard Sync:** Changes saved on this record page **automatically update the main dashboard** metrics.

The system dashboard parameters dynamically refresh to compute and reflect active changes across monthly tracking timelines, confirming successful end-to-end processing validation:



- **Automation Testing:** Validates that the 3-day reminder flow runs successfully and confirmation templates clear fields correctly.
- **Security Testing:** Reviews cross-team permissions to ensure profile access controls remain active across tables.
- **Data Validation:** Checks character constraints on phone entries, text string validation patterns, and data loading paths.
- **Dashboard Testing:** Verifies chart components refresh correctly and display real-time business metrics accurately.
- **Visual Metric Representation:** Renders the source data cleanly using an **interactive donut chart component** that shows the distribution of upcoming events at a glance.

- **Granular Time Segmentation:** The overview component breaks down total record counts dynamically by monthly targets, highlighting allocations for both **October 2026 and November 2026**.
- **On-Demand Refresh Capability:** Features a dedicated manual **Refresh** action bar control so managers can instantly query the latest backend database saves.

APPENDIX

System Component 1: Salesforce Environment Configuration (Milestone 1)

- **Action:** Created and activated a standalone **Salesforce Developer Edition** organization environment via the official signup engine.
- **Setup:** Configured custom administrative attributes and localized college/company parameters within the global setup profile.

System Component 2: Database Table Architecture (Milestone 2)

- **Action:** Built a custom database schema consisting of five key management objects and one background bridge table:
 - **Event__c:** Core data hub storing item dates, picklist categories, and transaction flags.
 - **Client__c:** Central repository tracking complete individual client records.
 - **Vendor__c:** Master profile listing vendor categories and standard business states.

- Venue__c: Location storage record mapping asset locations, capacities, and statuses.
- Feedback__c: Automated rating index utilizing an auto-numbering record generator (F-{{0000}}).
- EventVendor__c: Dedicated bridge table explicitly establishing a many-to-many lookup mesh between entities.

System Component 3: Custom Application Interface & Layouts (Milestone 3)

- **Action:** Deployed a series of custom application tabs to match the primary business objects.
- **UI Bundle:** Configured individual custom tab styles (such as building blocks, flags, and lightning icons) to match standard Salesforce workspace layouts.

System Component 4: Advanced Schema Fields & Formulas (Milestone 4)

- **Action:** Formatted field-level picklist variables and relational model definitions:
 - Configured specific lookup paths from **Event to Client** and **Event to Venue**.
 - Deployed a master-detail lookup hierarchy from the junction table directly to parent records.
 - Wrote an automated currency formula field using a CASE statement to auto-compute budgets according to custom event types:

System Component 5: Data Entry Integrity & Validation (Milestone 5)

- **Action:** Implemented programmatic input verification layers across data collection forms.
- **Logic:** Applied the `Email_Valid_Address` validation rule to the customer object utilizing a strict REGEX validation pattern to completely block unstructured or broken client text entries:

System Component 6: Manager Approval Processes (Milestone 6)

- **Action:** Developed an automated classic email framework paired with a multi-step manager workflow layout titled `Event_Cancellation_Process`.
- **Logic:** Evaluates transaction states so that when a record changes to Pending Cancellation, the system immediately locks the database entry, fires email alerts to coordinators, and either changes states to Canceled or resets it back to Confirmed based on the manager's review.

System Component 7: Declarative Process Flows (Milestone 7)

- **Action:** Configured a background record-triggered automation script titled `Client Reminder - 3 Days Before`.
- **Logic:** Runs via an action-optimized schedule path that tracks the event date, executing a text template engine exactly **3 days prior to the date field** to handle reminder alerts with owner carbon-copy (CC) routing.

System Component 8: Synchronous Automation Triggers (Milestone 8)

- **Action:** Programmed a programmatic back-end engine to handle internal platform records.
- **Logic:** Wrote an after insert and after update database class helper hook titled EventTrigger13 to communicate with a VenueStatusHelper class, automatically toggling venue states to Reserved or Available.

System Component 9: Schedule Conflict Prevention (Milestone 9)

- **Action:** Developed a custom back-end apex script mechanism labeled PreventDoubleBooking.
- **Logic:** Intercepts database save operations before execution, scanning for any existing bookings matching the target location and date, and invoking .addError() to fully stop conflicts at runtime.

System Component 10: Asynchronous Maintenance Batches (Milestone 10)

- **Action:** Programmed a daily cleanup framework using an asynchronous Database.Batchable execution model.
- **Logic:** Pairs a BatchCompleteEvents background cleaner class with a ScheduleCompleteEvents scheduling script, using the Salesforce central dashboard clock to automatically run every single evening at **8:00 PM** to change past data records to Completed.

System Component 11: Lightning Experience Branding (Milestone 11)

- **Action:** Packed all active metadata assets, tabs, and layout tables into a clean, custom-branded interface titled the **Event Planner App**.

- **Setup:** Customized unique branding assets, default logo imagery, security profiles, and standard toolbar navigation items.

System Component 12: Analytics Reporting Folders (Milestone 12)

- **Action:** Configured shared analytical workspaces titled *EventForce Reports* with granular view access controls mapped across company role titles.
- **Outputs:** Formatted summary matrices tracking entry metrics by calendar month using live column data fields.

System Component 13: Operational KPI Dashboards (Milestone 13)

- **Action:** Built a unified analytics center titled the *EventForce Operations Dashboard*.
- **Setup:** Linked data tables to live summary charts (such as Donut Charts), rendering real-time business performance summaries for system users.

System Component 14: Role Access & Profile Security (Milestones 14 - 18)

Action: Enforced an enterprise-grade security layer across all objects and user directories:

- Set organization-wide baseline field visibility policies entirely to **Private**.
- Cloned and managed custom security permission boundaries across specific business groups (*Admin*, *Coordinator*, *Vendor Manager*, and *Client Profile*).
- Configured a custom criterion-based data sharing rule labeled *Event_Sharing_For_Vendors* to safely open access to specific user paths.

FINAL OVERALL PROJECT FLOW

[PHASE 1: DISCOVERY & DATA PLANNING]

└─ Define Business Objectives → Draft Problem Matrix → Map Object Field Schemas

|



[PHASE 2: WORKSPACE SETUP & ENTITY RELATIONSHIPS]

└─ Spin up Dev Org → Deploy Custom Tables → Configure Lookup/Junction Matrix Models

|



[PHASE 3: CUSTOMIZATION LAYER & DATA RULES]

└─ Embed REGEX Check Rules → Build Budget Formulas → Script Custom Workspace Tabs

|



[PHASE 4: DISCRETE PROCESS AUTOMATIONS]

└─ Map Multi-Tier Email Alerts → Activate Cancellation Process Workflows

|



[PHASE 5: BACKGROUND FLOW CHANNELS]

└─ Build Record-Triggered Flows → Deploy Scheduled 3-Day Action Alert Paths

|



[PHASE 6: CUSTOM APEX ENGINES]

└─ Code Venue Asset Helpers → Embed Double-Booking Block Triggers → Deploy Nightly Batches

|



[PHASE 7: PROFILE-BASED SECURITY MESH]

└─ Clamp OWD to Private → Format Custom User Profiles → Build Cross-Team File Sharing Rules

|



[PHASE 8: PRESENTATION & OPERATIONAL SIGN-OFF]

└─ Package components into Lightning App → Map Monthly Reports → Deploy KPI Analytics Dashboards

Project Title: Event Force Management System

Platform: Salesforce CRM

Environment: Salesforce Developer Edition

Application Type: Event Management CRM

Primary Users: Event Coordinators, Event Administrators, Vendor Managers, Venue Managers, and Clients

